Title

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Abstract

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1 Introduction

1.1 Preliminaries

This subsection collects the notions on graphs and non-negative matrices used for both models, following the course slides [2, 3, 1].

1.1.1 Graphs

A *weighted digraph* $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$ has node set $V = \{1, 2, \dots, n\}$, edge set $E \subseteq V \times V$ and a weight $a_{ij} > 0$ for each edge $(i, j) \in E$; an edge (i, i) is a *self-loop*. Throughout the report the nodes are agents, and an edge (i, j) means that agent i is influenced by agent j. This is the convention of the course slides [1]; the tutorial by Proskurnikov and Tempo draws the edges in the opposite direction [4].

Definition 1.1 (Weighted adjacency matrix [2])**.** Given a weighted digraph $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$ of order n, its *weighted adjacency matrix* is the $n \times n$ matrix A whose (i, j) entry is equal to the edge weight a_{ij} if $(i, j) \in E$, and is equal to zero otherwise.

Definition 1.2 (Strong connectivity and globally reachable nodes [2])**.** A digraph $G = (V, E)$ is *strongly connected* if there is a directed walk from any node to any other node. A node of G is *globally reachable* if it can be reached from any other node through a directed walk.

With our convention, a directed walk from i to j means that agent i is influenced by agent j, either directly or through intermediaries. Hence G is strongly connected when every agent influences every other agent, at least indirectly, and a globally reachable node is an agent that influences the whole group.

Definition 1.3 (Periodic digraph [2])**.** A strongly connected digraph is *periodic* if there exists an integer $k > 1$ (the period) that divides the length of every simple cycle of the digraph; otherwise it is *aperiodic*.

Equivalently, a strongly connected digraph G is periodic if and only if the greatest common divisor of the lengths of its simple cycles is larger than 1 [2]. In particular, a strongly connected digraph with a self-loop is aperiodic, since the self-loop is a cycle of length 1.

Definition 1.4 (Strongly connected components and condensation digraph [1])**.** Given a weighted digraph G, a subgraph S is a *strongly connected component* of G if S is strongly connected and any other subgraph of G strictly containing S is not strongly connected. The *condensation digraph* $C(G)$ is obtained by collapsing each strongly connected component of G into a single node: for any two distinct strongly connected components S_k and S_h, there is an edge (S_k, S_h) in $C(G)$ if and only if there exists an edge $(i, j) \in E$ with $i \in S_k$ and $j \in S_h$ (self-loops are not allowed in $C(G)$). A node of $C(G)$ with no outgoing edges is called a *sink* [2].

A sink of $C(G)$ is a group of agents that are influenced only by members of the group itself: no agent outside the group can change their opinions.

If G contains a globally reachable node, then $C(G)$ has a unique sink [1].

1.1.2 Matrices

Definition 1.5 (Row-stochastic matrix [3])**.** A matrix $A \in \mathbb{R}^{n\times n}$ is *row-stochastic* if it is non-negative and $A\mathbf{1}_n = \mathbf{1}_n$, i.e. each row sums to 1.

Theorem 1.6 (Spectral properties of row-stochastic matrices [3])**.** *If $A \in \mathbb{R}^{n\times n}$ is row-stochastic, then $1 \in \operatorname{spec}(A)$ and $\rho(A) = 1$; hence $\operatorname{spec}(A)$ is contained in the closed unit disk.*

Definition 1.7 (Semi-convergent matrix [3])**.** A matrix $A \in \mathbb{R}^{n\times n}$ is *semi-convergent* if $\lim_{k\to+\infty} A^k$ exists and is finite.

The properties of a non-negative matrix that determine its spectrum are characterized by the graph notions above.

Definition 1.8 (Irreducible and primitive matrices [3])**.** A non-negative matrix $A \in \mathbb{R}^{n\times n}$, $n \geq 2$, is *irreducible* if $\sum_{k=0}^{n-1} A^k$ is positive, and *primitive* if there exists $k \in \mathbb{N}$ such that A^k is positive.

Theorem 1.9 (Irreducible matrices and digraphs [2])**.** *Let G be a weighted digraph of order $n \geq 2$ with weighted adjacency matrix A. Then A is irreducible if and only if G is strongly connected.*

Theorem 1.10 (Primitive matrices and digraphs [2])**.** *Let G be a weighted digraph of order $n \geq 2$ with weighted adjacency matrix A. Then A is primitive if and only if G is strongly connected and aperiodic. In particular, if G is strongly connected and has at least one self-loop, then A is primitive.*

2 FDG

2.1 Introduction to the FDG model

2.1.1 The FDG model

Consider a group of individuals $V = \{1, 2, \ldots, n\}$, where $x_i \in \mathbb{R}$ is the opinion that individual i holds on a given topic. The influences among the individuals are described by a weighted digraph $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$: following the convention of Section 1.1, an edge $(i, j) \in E$, with weight $a_{ij} > 0$, means that individual i is influenced by individual j.

The French–Harary–DeGroot (FDG) model [1], called French–DeGroot model in [4], describes how each individual, or agent, updates its opinion: the new opinion is a weighted average of the opinions of the agents it is influenced by,

$$x_i(k+1) = \sum_{j=1}^{n} a_{ij}\, x_j(k), \qquad \sum_{j=1}^{n} a_{ij} = 1, \tag{1}$$

or, in vector form, $x(k+1) = Ax(k)$, where A is the weighted adjacency matrix of G (Definition 1.1). The constraint on the weights means that A is row-stochastic (Definition 1.5).

Since $x(k) = A^k x(0)$, the asymptotic behaviour of the model is determined by the powers of A. Three outcomes are possible:

- *no convergence*: $\lim_{k\to+\infty} x(k)$ does not exist for some $x(0)$; this happens when A is not semi-convergent (Definition 1.7), e.g. when A has an eigenvalue $\mu \neq 1$ with $|\mu| = 1$;
- *consensus*: $\lim_{k\to+\infty} x(k) = \alpha \mathbf{1}_n$ for some $\alpha \in \mathbb{R}$;
- *disagreement*: $\lim_{k\to+\infty} x(k) = v$ with $v \neq \alpha \mathbf{1}_n$ for every $\alpha \in \mathbb{R}$.

2.1.2 Consensus and social power

The following theorem characterizes when the FDG model reaches a consensus. Its condition (A3) involves the subgraph of globally reachable nodes: this set of nodes is always strongly connected [1]: a walk between two globally reachable nodes only visits nodes that are themselves globally reachable. Hence its periodicity is well defined (Definition 1.3).

Theorem 2.1 (Consensus in DT averaging systems [1])**.** *Let G be a weighted digraph with row-stochastic adjacency matrix A. The following statements are equivalent:*

(A1) the eigenvalue 1 is simple and all other eigenvalues μ satisfy $|\mu| < 1$;

(A2) A is semi-convergent and $\lim_{k\to+\infty} A^k = \mathbf{1}_n w^\top$, where $w \in \mathbb{R}^n$ satisfies $w \geq 0$, $\mathbf{1}_n^\top w = 1$ and $w^\top A = w^\top$;

(A3) G contains a globally reachable node and the subgraph of globally reachable nodes is aperiodic.

If any, and therefore all, of the previous conditions are satisfied, then

(i) w is the left eigenvector of A associated with the eigenvalue 1, and $w_i > 0$ if and only if node i is globally reachable;

(ii) the solution of the FDG model satisfies $\lim_{k\to+\infty} x(k) = \left(w^\top x(0)\right) \mathbf{1}_n$.

Corollary 2.2 (Average consensus [1])**.** *If A is doubly-stochastic, i.e. row-stochastic with also $\mathbf{1}_n^\top A = \mathbf{1}_n^\top$, and (A3) holds, then $w = \frac{1}{n}\mathbf{1}_n$ and $\lim_{k\to+\infty} x(k) = \left(\frac{1}{n}\sum_{i=1}^n x_i(0)\right) \mathbf{1}_n$.*

The entry w_i is called the *social power* of agent i: it is the weight of the initial opinion $x_i(0)$ in the final opinion of the group [4].

Point (i) has a direct social interpretation. Since the consensus value $w^\top x(0)$ is the same for every agent, the initial opinion of agent i can contribute to it only if it reaches *all* the agents. If i is not globally reachable, there is at least one agent that is influenced by i neither directly nor through intermediaries: the opinion of i never reaches that agent, so it cannot enter the common value, and $w_i = 0$. Conversely, the opinion of a globally reachable agent reaches the whole group and has a positive weight. Hence the consensus value depends only on the initial opinions of the globally reachable agents.

2.1.3 Disagreement

Theorem 2.1 requires a single globally reachable group, i.e. a unique sink in the condensation digraph $C(G)$. We now consider the case in which $C(G)$ has $n_s \geq 2$ sinks. Each sink is a strongly connected component, so its periodicity is well defined (Definition 1.3).

Theorem 2.3 (Disagreement in DT averaging systems [1])**.** *Let G be a weighted digraph with row-stochastic adjacency matrix A. Assume that the condensation digraph $C(G)$ has $n_s \geq 2$ sinks. The following statements are equivalent:*

(A1) the eigenvalue 1 is semi-simple with multiplicity n_s and all other eigenvalues μ satisfy $|\mu| < 1$;

(A2) A is semi-convergent;

(A3) each sink of $C(G)$ is aperiodic.

If any, and therefore all, of the previous conditions are satisfied, then the solution of the FDG model satisfies $\lim_{k\to+\infty} x(k) = v \neq \alpha\mathbf{1}_n$ for every $\alpha \in \mathbb{R}$.

The final opinions can be described group by group. The agents of a sink are not influenced by anyone outside the sink, so each sink evolves as an FDG model on its own and, being aperiodic, reaches an internal consensus that depends only on the initial opinions of its members. Every agent outside the sinks converges instead to a convex combination of the consensus values of the sinks [1].

In social terms, a sink is a closed group that listens only to itself, and two closed groups never influence each other: each of them keeps its own consensus value. The agents in between listen, directly or through intermediaries, to several closed groups and settle on an intermediate opinion, closer to the groups they listen to more. As a consequence, the disagreement is bounded: every final opinion lies between the smallest and the largest consensus value of the sinks.

2.1.4 Rosetta stone

The results of this section, together with the consensus and disagreement theorems above, classify all the possible behaviours of the FDG model. The classification is summarized in Table 1, the so-called *Rosetta stone* [1], which translates each property of the row-stochastic matrix A into a property of the digraph G and into the corresponding asymptotic outcome; for each case, the table also reports an example digraph, its matrix and its spectrum.

The two consensus cases illustrate Theorem 2.1. In case (i) both agents are globally reachable and A is doubly-stochastic, so by Corollary 2.2 $w = (1/2, 1/2)^\top$ and the group agrees on the average $\big(x_1(0) + x_2(0)\big)/2$. In case (iii) agent 2 is influenced by agent 1 but not vice versa, so only agent 1 is globally reachable: $w = (1, 0)^\top$ and the group agrees on $x_1(0)$, whatever the initial opinion of agent 2.

Case (iv) illustrates Theorem 2.3 in its simplest form: each agent is influenced only by itself, so each agent is a sink of $C(G)$ ($n_s = 2$), both sinks are aperiodic thanks to the self-loops, and every agent keeps its initial opinion, $x(k) = x(0)$ for all k.

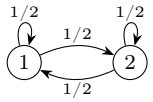
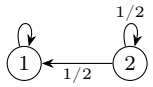
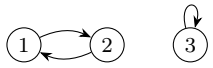
Case	Example digraph	A	spec(A)	Property of A / of G	Outcome
(i)		$\begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$	{1, 0}	A primitive / G strongly connected and aperiodic	consensus
(ii)		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	{1, −1}	A irreducible, not primitive / G strongly connected and periodic	no convergence
(iii)		$\begin{bmatrix} 1 & 0 \\ 1/2 & 1/2 \end{bmatrix}$	{1, 1/2}	1 simple and strictly dominant, A not primitive / $C(G)$ has a unique sink, which is aperiodic	consensus
(iv)		$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	{1, 1}	1 semi-simple but not simple, strictly dominant / $C(G)$ has at least two sinks, all aperiodic	disagreement
(v)		$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	{1, 1, −1}	1 not strictly dominant / G not strongly connected, at least one sink of $C(G)$ is periodic	no convergence

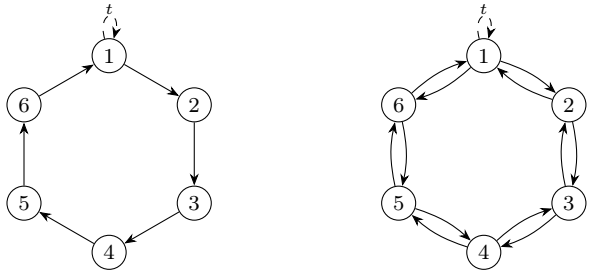
Table 1: Rosetta stone of the FDG model $x(k+1) = Ax(k)$, with A row-stochastic: for each case, an example digraph (unlabelled edges have weight 1), its matrix A, the spectrum of A, the general property of A and of G, and the outcome. Cases (i)–(iv) and the examples follow [1]; the properties of case (v), left blank in the slides, follow from [4].

2.2 Stubbornness and social power in the FDG model

In the FDG model the simplest way to account for stubbornness is the self-weight a_{ii}, i.e. the weight of the self-loop of agent i. This section studies how self-weights, and stubborn agents in particular, affect the two aspects introduced in Section 2.1: whether the group reaches an agreement, and how the social power is distributed among the agents.

All the examples use the two networks of Figure 1, both with $n = 6$ agents. In the *directed cycle* each agent is influenced only by the next one, $a_{i,i+1} = 1$; in the *bidirectional ring* each agent is influenced by its two neighbours with equal weights, $a_{i,i-1} = a_{i,i+1} = 1/2$ (indices modulo 6). Socially, in the first network everyone listens to a single person, while in the second

everyone listens to the two people next to them. Without self-loops both digraphs are strongly connected and periodic: the directed cycle has a single cycle, of length 6, while every cycle of the bidirectional ring has even length, so its period is 2. Hence neither network reaches an agreement (case (ii) of Table 1). Moreover, both matrices are doubly-stochastic, so $w = \frac{1}{6}\mathbf{1}_6$ is the left eigenvector associated with the eigenvalue 1: no agent is favoured by the structure of the network, and any asymmetry in the social power will be due only to the self-weight added to agent 1.



(a) directed cycle (b) bidirectional ring

Figure 1: The two networks used in Sections 2.2 and 3. An edge $i \to j$ means that agent i is influenced by agent j. In (a) all edges have weight 1, in (b) all edges have weight 1/2. The dashed self-loop of agent 1, of weight t, is the only modification considered: the other edges of agent 1 are scaled by $1 - t$ (equation (2)).

2.2.1 Modelling stubbornness: self-weights and stubborn agents

The self-weight a_{ii} is the weight that agent i gives to its own *current* opinion; it indicates how open the agent is to the opinions of the others [4]. An agent with $a_{ii} = 0$ is open-minded and relies completely on the others; an agent with a positive self-weight changes its opinion more slowly, but it still moves towards the opinions it listens to. In social terms, a self-weight is a form of inertia, not an anchor to a fixed opinion.

Definition 2.4 (Stubbornness [4])**.** An agent is *stubborn* if its opinion remains unchanged independently of the others' opinions. In the FDG model, agent i is stubborn, i.e. $x_i(k) = x_i(0)$ for all k, if and only if $a_{ii} = 1$ (and hence $a_{ij} = 0$ for all $j \neq i$).

Stubbornness is therefore the extreme case of the self-weight, and there is no intermediate degree of it: an agent with $0 < a_{ii} < 1$ is not stubborn, it only has a positive self-weight.

In the digraph, a stubborn agent has no edges towards the other agents: it is influenced by nobody, while the agents that listen to it keep doing so. Hence it forms a strongly connected component made of a single node, which is a sink of $C(G)$ and is aperiodic because of its self-loop. If, in addition, every agent is influenced by the stubborn agent i, directly or through intermediaries, then i is the only globally reachable node, and Theorem 2.1 gives $w = e_i$: the whole group converges to the initial opinion of the stubborn agent (see also [4]).

To move continuously from no self-weight to stubbornness, we modify only the row of agent 1. Given a row-stochastic matrix A with $a_{11} = 0$, as in both networks of Figure 1, we replace its first row by

$$a_1(t)^\top = (1 - t)\, a_1^\top + t\, e_1^\top, \qquad t \in [0, 1], \tag{2}$$

where e_1 is the first vector of the canonical basis, and leave the other rows unchanged. Agent 1 gives weight t to itself and splits the remaining weight $1 - t$ among the agents it listened to before, in the same proportions. The parameter t is thus the self-weight a_{11}: $t = 0$ is the original network, $0 < t < 1$ gives agent 1 a positive self-weight, and $t = 1$ makes agent 1 stubborn. For the directed cycle the first row becomes $(t,\, 1-t,\, 0,\, 0,\, 0,\, 0)$; for the bidirectional ring it becomes $(t,\, \frac{1-t}{2},\, 0,\, 0,\, 0,\, \frac{1-t}{2})$. In the following we consider the three values $t = 0$, $t = 1/2$ and $t = 1$.

2.2.2 Self-weights, stubbornness and convergence

The three values of t correspond to three different cases of the Rosetta stone.

- $t = 0$: both digraphs are strongly connected and periodic, so neither network converges (case (ii)).
- $0 < t < 1$: agent 1 still listens to its neighbours with total weight $1 - t > 0$, so the digraph stays strongly connected, and the self-loop of agent 1 is a cycle of length 1, so the digraph becomes aperiodic. By Theorem 1.10 the matrix is primitive, and the group reaches a consensus (case (i)).
- $t = 1$: agent 1 is stubborn. By the discussion in Section 2.2.1, it is the only sink of $C(G)$ and the only globally reachable node, so the group reaches a consensus on its initial opinion $x_1(0)$ (case (iii)).

The same picture can be read on the spectrum of the matrix. For the directed cycle the characteristic polynomial is

$$p(\lambda) = (\lambda - 1)\left[\lambda^5 + (1-t)(\lambda^4 + \lambda^3 + \lambda^2 + \lambda + 1)\right],$$

Apart from $\lambda_1 = 1$, the eigenvalues are the roots of a polynomial of degree 5, which for a generic t have no closed-form expression; for the three values of t they are

- $t = 0$: $\lambda_1 = 1$, $\lambda_{2,3} = \frac{1}{2} \pm \frac{\sqrt{3}}{2}\, i$, $\lambda_{4,5} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}\, i$, $\lambda_6 = -1$ (the sixth roots of unity, all on the unit circle);
- $t = 1/2$: $\lambda_1 = 1$, $\lambda_{2,3} \approx 0.535 \pm 0.751\, i$, $\lambda_{4,5} \approx -0.373 \pm 0.759\, i$, $\lambda_6 \approx -0.823$;
- $t = 1$: $\lambda_1 = 1$, $\lambda_2 = \cdots = \lambda_6 = 0$.

For the bidirectional ring the characteristic polynomial is

$$p(\lambda) = (\lambda - 1)\big(\lambda - \tfrac{1}{2}\big)\big(\lambda + \tfrac{1}{2}\big)\Big[\lambda^3 + (1-t)\lambda^2 - \tfrac{2t+1}{4}\,\lambda + \tfrac{t-1}{4}\Big],$$

The eigenvalues 1 and $\pm\frac{1}{2}$ do not depend on t, while the other three are the roots of the cubic factor, whose closed-form expression in t is too involved to be informative; for the three values of t the eigenvalues are

- $t = 0$: $\lambda_1 = 1$, $\lambda_{2,3} = \frac{1}{2}$, $\lambda_{4,5} = -\frac{1}{2}$, $\lambda_6 = -1$ (on the unit circle);
- $t = 1/2$: $\lambda_1 = 1$, $\lambda_2 \approx 0.623$, $\lambda_3 = \frac{1}{2}$, $\lambda_4 \approx -0.223$, $\lambda_5 = -\frac{1}{2}$, $\lambda_6 \approx -0.901$;
- $t = 1$: $\lambda_1 = 1$, $\lambda_2 = \frac{\sqrt{3}}{2}$, $\lambda_3 = \frac{1}{2}$, $\lambda_4 = 0$, $\lambda_5 = -\frac{1}{2}$, $\lambda_6 = -\frac{\sqrt{3}}{2}$.

In both networks, for $t = 0$ some eigenvalues different from 1 lie on the unit circle, and they cause the oscillations; for $t > 0$ all the eigenvalues different from 1 lie strictly inside the unit disk, and the group converges. For $t = 1$ the directed cycle is left with the eigenvalue 0 only, which is why its convergence is reached in a finite number of steps.

Figure 2 shows the opinions $x_i(k)$ in the six cases. For $t = 0$ the opinions keep rotating along the directed cycle and keep oscillating between two configurations on the bidirectional ring. For $t = 1/2$ both networks reach a consensus whose value is a weighted average of all the initial opinions, with weights given by the social power (Section 2.2.3). For $t = 1$ both networks reach a consensus on $x_1(0)$, but in different ways. In the directed cycle every agent other than agent 1 simply copies the opinion of the next agent, so the opinion of agent 1 travels backwards along the chain, one agent per step, and the consensus is reached exactly after $n - 1 = 5$ steps. In

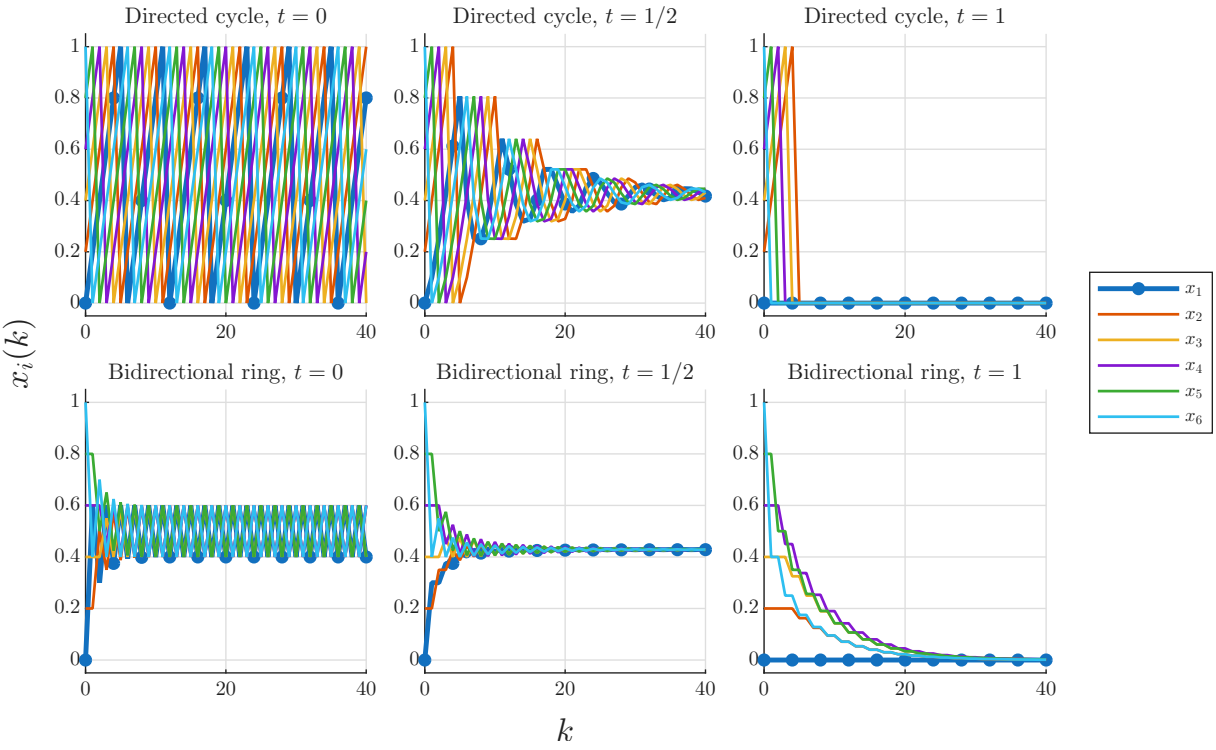


Figure 2: Opinions $x_i(k)$ in the two networks of Figure 1 (top: directed cycle; bottom: bidirectional ring) for three values of the self-weight t of agent 1 (columns), with $x(0) = (0, 0.2, 0.4, 0.6, 0.8, 1)^\top$. Agent 1 is marked with dots.

the bidirectional ring every agent averages the opinions of its two neighbours, so the opinions approach $x_1(0)$ gradually, the neighbours of agent 1 first.

In social terms, a group in which nobody gives weight to its own opinion can keep passing opinions around forever. It is enough that a single agent gives some weight to its current opinion, without being stubborn, for the whole group to reach an agreement. If that agent becomes stubborn, the agreement is reached on its own opinion. The structure of the network decides only how this happens: by copying along a chain in the directed cycle, by averaging with the neighbours in the bidirectional ring.

2.2.3 Self-weights, stubbornness and social power

For $t > 0$ both networks reach a consensus (Section 2.2.2), so the social power w of Section 2.1.2 is well defined, and Theorem 2.1 describes how it depends on t.

- $0 < t < 1$: the digraph is strongly connected, so every agent is globally reachable and, by point (i) of Theorem 2.1, every agent has a positive social power. Agent 1 gives weight to its own opinion, but it does not take all the influence: the initial opinion of every agent enters the consensus value.

- $t = 1$: agent 1 is stubborn and is the only globally reachable node, so $w = e_1$: the stubborn agent takes all the social power, and the initial opinions of the other agents have no effect on the final opinion of the group.

Between these two extremes, the social power can be computed as the left eigenvector of $A(t)$ associated with the eigenvalue 1, normalized so that its entries sum to 1 (condition (A2) of Theorem 2.1). Figure 3 reports the result for $t \in (0, 1]$. Numerically, the social power of agent 1 grows with t towards 1, while the other agents share the remaining social power equally; for $t = 1/2$, agent 1 has social power ≈ 0.286 and every other agent ≈ 0.143. Moreover, the

computation gives the same values on the two networks. In other words, on these two networks the self-weight of agent 1 changes how the opinions converge, which depends on the network (Figure 2), but not where they converge.

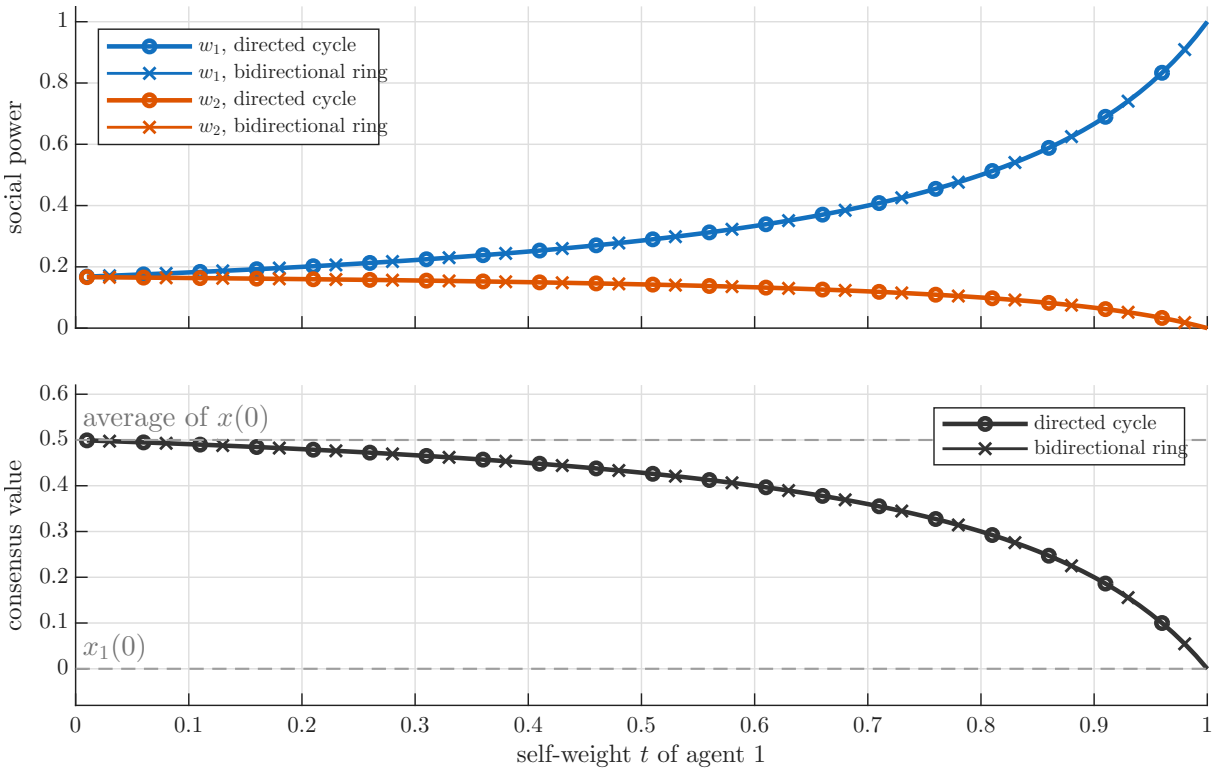


Figure 3: Social power of agent 1 (w_1) and of agent 2 (w_2) as a function of the self-weight $t \in (0, 1]$ of agent 1, computed numerically in the directed cycle (dots) and in the bidirectional ring (crosses); the two networks give the same values.

Remark 2.5 (More than one stubborn agent). If two or more agents are stubborn, each of them is a sink of $C(G)$ made of a single, aperiodic node. If every other agent is influenced, directly or through intermediaries, by at least one stubborn agent, these are the only sinks, and by Theorem 2.3 the group converges, but to a disagreement: every stubborn agent keeps its own initial opinion, and every other agent converges to a convex combination of the opinions of the stubborn agents it is influenced by (see also [4]). Since there is no single consensus value, the social power is no longer defined. This situation, in which several agents are anchored to different fixed opinions, is the starting point of the Friedkin–Johnsen model of Section 3.

3 FJ

3.1 Introduction to the FJ model

3.2 Stubborness and social power in the FJ model

4 Conclusions

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